

Riley Beenders

MECHANICAL DESIGN | DESIGN COORDINATION | ENGINEERING LEADERSHIP

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PROFESSIONAL SUMMARY

Mechanical and electromechanical engineering leader with 5+ years of professional engineering experience and 10+ years of multidisciplinary project work. Owns complex mechanical scopes from concept design and drawings through vendor coordination, procurement, fabrication, assembly, and field support, with hands-on regulatory compliance work, engineering hiring, and consensus-building across disciplines.

CORE EXPERTISE

Mechanical Design: SolidWorks, Autodesk Inventor, Siemens NX, CATIA, and Fusion 360; GD&T; design for manufacturing and assembly (DFM/DFA); fixture and tooling design; thermal and acoustic enclosure design; motion and drive systems

Project Delivery: Design coordination with vendors and fabricators; procurement, sourcing, and logistics; engineering documentation and BOM control; vendor communication and cost negotiation; field troubleshooting and support

Compliance & Leadership: Product safety and regulatory compliance (2006/42/EC machinery directive, EU and U.S. standards); risk analysis; technical writing; hiring and technical interviews; worldwide project collaboration

PROFESSIONAL EXPERIENCE

Proteor / Proteor Printing Solutions | Coplay, PA

January 2025 - Present

Lead, Research & Development Engineer

- Owned the mechanical, electrical, and software design of ICARUS-Lite from concept through commercialization, including a bolt-less shell, custom AC heaters, and product changes supporting 2006/42/EC machinery directive conformity.
- Re-engineered POSEIDON and ICARUS with a standardized linear-motor architecture and swappable mounts, demonstrating 4-micron precision at 4 m/s with a 30 kg moving mass while reducing support complexity.
- Co-developed a custom hotend with an external manufacturing partner, improving reliability and reducing cost, and performed risk analysis and mitigations to EU and U.S. standards.
- Act as Hiring Manager for engineering roles and provide cross-functional technical support for the product line, from troubleshooting to after-sales service.

Filament Innovations | Whitehall, PA

May 2020 - January 2025

Chief Technology Officer (Jan 2023 - Jan 2025) | Lead Electro-Mechanical Engineer (Jun 2022 - Jan 2023) | Industrial Engineer (Apr 2021 - Jun 2022) | Engineering Technician (May 2020 - Apr 2021)

- Led the design and launch of Gen3 POSEIDON, combining a 1 m³ build volume, braked NEMA 34 Z drives, industrial-servo XY motion, and a custom gantry.
- Designed THE KRAKEN, a 1 x 2 x 1.5 m gantry system, and managed its CAD, BOM, sourcing, fabrication, assembly, firmware tuning, customer support, and operator training.
- Improved thermal and acoustic performance through enclosure redesign and component placement, including custom tooling developed with local fabricators for large-radius aluminum-panel bends.
- Coordinated custom harnesses, panel designs, and hardware improvements with vendors, and expanded motion safety with planetary gearboxes and power-failure brakes.

ADDITIONAL RELEVANT EXPERIENCE

RJB.Engineering - Owner & Engineering Consultant | 2015 - Present | Deliver project-based engineering support for automotive clients and design and manufacture printer hardware accessories.

Blue Mountain Ski Resort - Ski Lift Operator | 2019 - 2021 | Performed motor and control safety checks on passenger lifts and supported maintenance during breakdowns while assisting thousands of guests.

EDUCATION & PROFESSIONAL DEVELOPMENT

Penn State University | B.S., Electro-Mechanical Engineering

December 2024

Relevant development: Penn State Certificate in Engineering Design and Tools (2023); Lafayette College Mechanical Engineering Certificate (2019); Lean Six Sigma Yellow Belt (2025); MathWorks Simscape Onramp (2026)